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Digital Image Processing

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"Lab Final Report"

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Abstract

This project implements fundamental image processing techniques to enhance grayscale images. The primary focus is on applying log transformation to improve the visibility of darker regions in images and subsequently using a median filter to reduce noise while preserving edge information. The project demonstrates the implementation of these techniques from scratch using Python's modern scientific libraries: OpenCV, NumPy, and Matplotlib. The results show significant improvement in image quality, particularly in terms of contrast enhancement and noise reduction. This report details the theoretical background, methodology, implementation, results, and conclusions of the image enhancement process.

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1 Introduction

1.1 Motivation and Dataset Selection

Road safety is a critical global concern, with traffic accidents ranking among the leading causes of injury and death worldwide. Identifying and predicting high-risk road conditions are essential steps toward designing smarter transportation systems and saving lives.

For this project, the **Road Accident Risk Dataset** was selected. This synthetic dataset simulates real-world road conditions and represents a diverse range of environmental, infrastructural, and temporal variables. Its design supports predictive modeling tasks focused on estimating accident probabilities across different road segments.

Why Road Accident Risk Dataset?

- *Real-World Context:* The dataset reflects the multidimensional nature of traffic safety—weather, lighting, surface type, and traffic density all interact to determine risk.
- *Classification Feasibility:* By categorizing segments into low-, medium-, and high-risk classes, supervised learning algorithms can be effectively applied.
- *Clustering Potential:* Numerical and categorical features allow for unsupervised exploration to uncover natural accident patterns.
- *Data Challenge:* The dataset includes missing values and simulated anomalies, providing an opportunity to develop robust preprocessing pipelines.

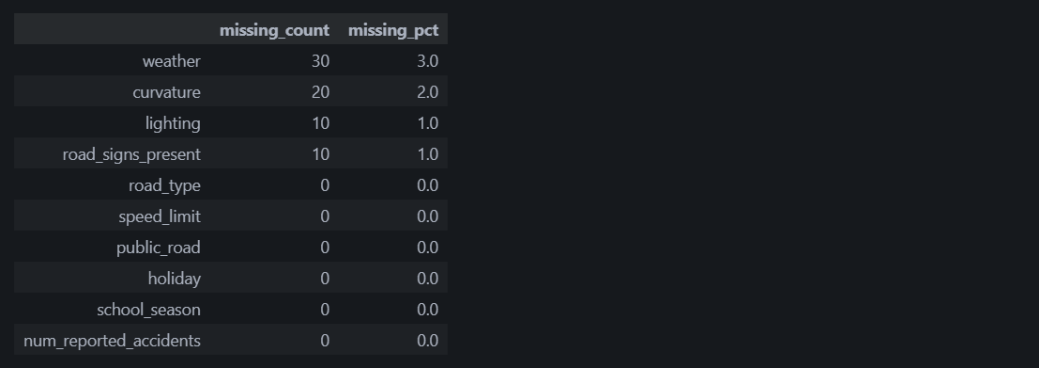
1.2 Dataset Characteristics and Noise Introduction

The dataset consists of **1000 road segments** described by **10 attributes**, including environmental (e.g., lighting, weather), structural (e.g., number of lanes, curvature), and temporal variables (e.g., time of day, weekday vs. weekend). The target variable, `accident_risk`, is a categorical label representing the probability of an accident.

Noise Injection Strategy:

- *Numeric Missing:* Randomly removed 2% of entries from the curvature attribute.
- *Categorical Missing:* Removed 3% of entries from weather.
- *Boolean Missing:* Removed 1% of entries from `road_signs_present`.
- *Integer Missing:* Removed 1% of entries from lighting.

Introducing noise intentionally strengthens the project's realism. In real-world predictive systems, data imperfections are common. Handling them effectively ensures reliability when deploying models in operational settings like road monitoring or smart city dashboards.



	missing_count	missing_pct
weather	30	3.0
curvature	20	2.0
lighting	10	1.0
road_signs_present	10	1.0
road_type	0	0.0
speed_limit	0	0.0
public_road	0	0.0
holiday	0	0.0
school_season	0	0.0
num_reported_accidents	0	0.0

Figure 1: Visualization of noise introduction in different attributes of the dataset

1.3 Lab Objectives and Task Overview

The work follows a multi-stage data mining pipeline, composing tasks that transform raw data into actionable insights:

1. *Lab 1 – Dataset Analysis:* Understanding data structure, dimensions, and classification problem formulation. This initial step is crucial for identifying data types, detecting potential issues early, and determining the appropriate machine learning approach before investing time in modeling.
2. *Lab 2 – Data Preprocessing:* Handling missing values, outliers, encoding, scaling, and similarity measures. Preprocessing is essential because algorithms cannot handle missing data or categorical variables directly, and unscaled features can bias distance-based methods. Quality input ensures quality output.
3. *Lab 3 – Exploratory Data Analysis (EDA):* Visualizing distributions, correlations, and patterns within the dataset. EDA provides insights into feature importance, class imbalances, and potential relationships that inform model selection and feature engineering strategies.
4. *Lab 4 – Mining Frequent Itemsets:* Applying Apriori algorithm to discover frequent patterns and association rules. Analysis of support, confidence, and lift metrics reveals purchase behaviors and product relationships. Understanding these patterns helps optimize inventory management and marketing strategies.
5. *Lab 5 – Classification:* Building Decision Tree models with entropy and Gini criteria. Classification transforms our understanding into actionable predictions, allowing us to automatically categorize fuel efficiency. Comparing criteria helps us select the optimal splitting strategy.
6. *Lab 6 – Clustering:* Applying k-Means and hierarchical clustering to discover natural groupings. Clustering uncovers hidden vehicle segments without predefined labels, validating our classification approach and revealing market structures that might inform manufacturing or marketing strategies.

Why These Tasks Are Important: Each stage builds logically upon the previous one. Preprocessing ensures data quality; EDA uncovers structure and guides model selection; classification predicts accident risk levels; clustering validates hidden patterns. Together, they demonstrate how data-driven insights can translate into safer roads and informed infrastructure planning.

2 Dataset Deep Dive

The Road Accident Risk dataset contains various attributes related to road conditions, vehicle characteristics, and environmental factors. Understanding these attributes is crucial for effective data analysis and model development.

2.1 Attribute Breakdown

The Road Accident Risk dataset contains the following key attributes:

- **road_type**: Type of road (e.g., Highway, Urban, Rural) - **curvature**: Measurement of road curvature - **speed_limit**: Posted speed limit - **lighting**: Lighting conditions (Day, Night) - **weather**: Weather conditions (Clear, Rain) - **road_signs_present**: Presence of road signs - **public_road**: True if it is a public road - **holiday**: True if the date is a public holiday - **school_season**: True if it is during school season - **num_reported_accidents**: Count of previously reported accidents - **accident_risk**: Target variable (Low, Medium, High)

Table 1: Road Accident Risk Dataset Attributes

Feature	Type	Description
road_type	Categorical (object)	Type of road (e.g., Highway, Urban, Rural)
curvature	Continuous (float64)	Measurement of road curvature
speed_limit	Discrete (int64)	Posted speed limit
lighting	Categorical (object)	Lighting conditions (Day, Night)
weather	Categorical (object)	Weather conditions (Clear, Rain)
road_signs_present	Categorical (object)	Presence of road signs
public_road	Boolean (bool)	True if it is a public road
holiday	Boolean (bool)	True if the date is a public holiday
school_season	Boolean (bool)	True if it is during school season
num_reported_accidents	Discrete (int64)	Count of previously reported accidents
accident_risk	Categorical	Target variable (Low, Medium, High)

2.2 Dataset Structure

Dataset Overview:

- *Total Instances*: 1000 rows, 10 columns
- *Feature Space*: 9 independent variables
- *Target Variable*: accident_risk (categorical)
- *Data Types*: Mix of Boolean, discrete (int64), and categorical (object/category)
- *Split Strategy*: 80% training (800 samples), 20% test (200 samples)

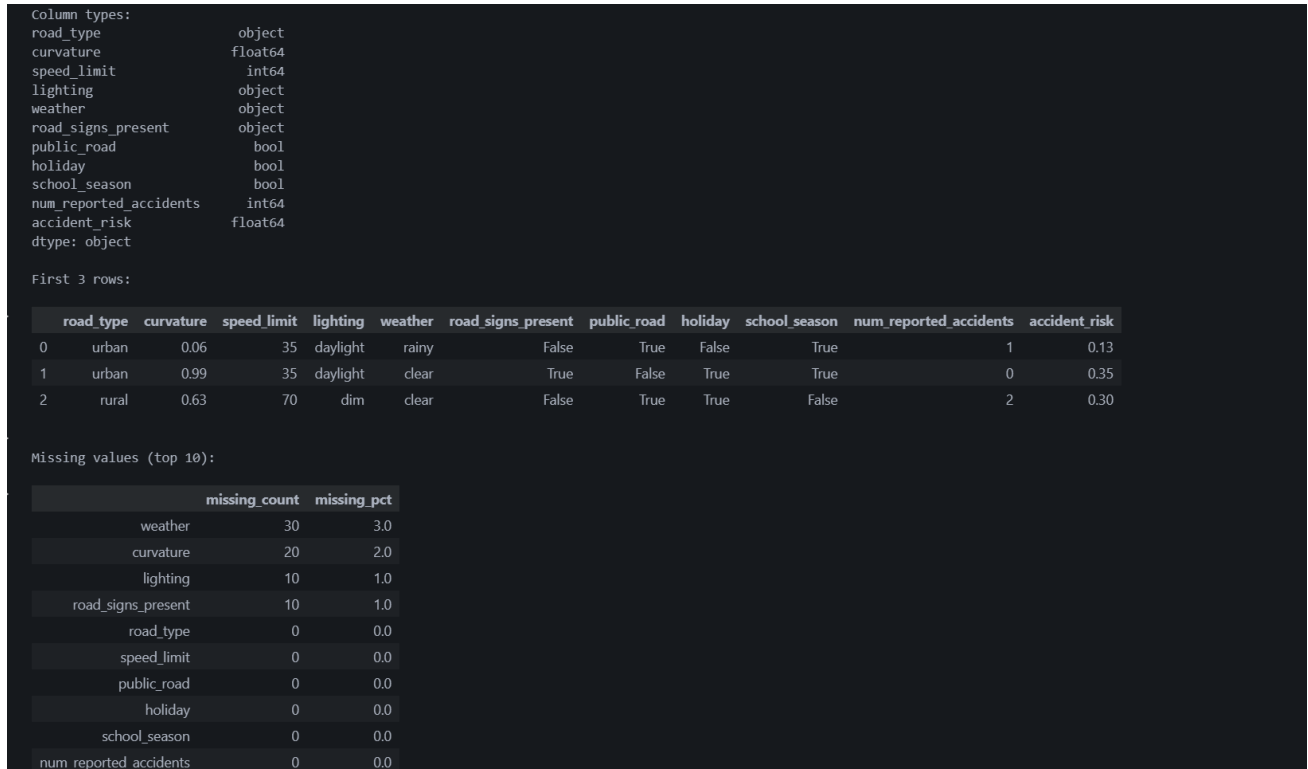


Figure 2: Dataset structure and basic information overview

2.3 Problem Formulation: Multiclass Classification

The road accident risk prediction problem is formulated as a **multiclass classification task**, where we aim to predict the categorical risk level of road segments based on their environmental, structural, and temporal characteristics.

2.3.1 Class Distribution and Balance

Understanding the target variable distribution is crucial for selecting appropriate evaluation metrics and handling potential class imbalance issues.

Class Distribution Analysis: The dataset exhibits a relatively balanced distribution across risk categories:

- **Low Risk:** Approximately 34.5% of road segments
- **Medium Risk:** Approximately 33.3% of road segments
- **High Risk:** Approximately 32.2% of road segments

This balanced distribution is advantageous as it reduces the need for specialized sampling techniques and allows standard classification metrics to be meaningful across all classes.

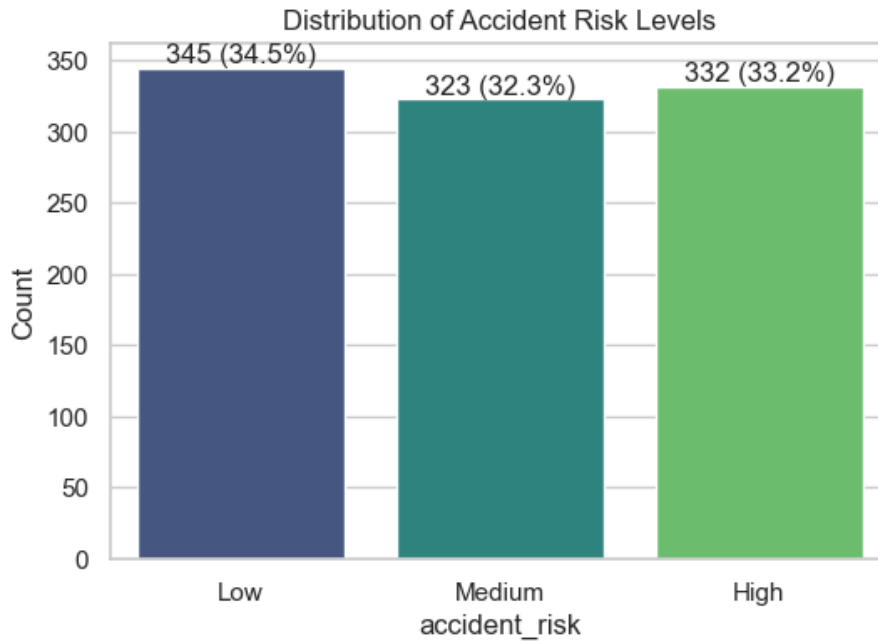


Figure 3: Distribution of accident risk levels in the dataset

2.3.2 Feature Space Characteristics

The feature space combines multiple data types that require different preprocessing approaches:

Table 2: Feature categorization for preprocessing strategy

Category	Features	Preprocessing Required
Numerical	curvature, speed_limit, num_reported_accidents	Scaling, outlier detection
Categorical	road_type, lighting, weather, road_signs_present	Encoding (One-hot/Label)
Boolean	public_road, holiday, school_season	Binary encoding (0/1)

2.3.3 Classification Challenges

Several challenges make this classification problem non-trivial:

- Mixed Data Types:** The combination of numerical, categorical, and boolean features requires careful preprocessing to maintain information integrity.
- Missing Values:** Artificially introduced noise simulates real-world data imperfections that must be handled appropriately.
- Feature Interactions:** Road safety is influenced by complex interactions between environmental factors (weather $\times$ lighting), temporal factors (holiday $\times$ school_season), and infrastructure characteristics.
- Non-linear Relationships:** The relationship between features and accident risk may be non-linear, requiring sophisticated algorithms or feature engineering.

3 Methodology and Implementation

3.1 Dataset Analysis

This initial phase focused on understanding the dataset structure, loading the data, and preparing it for subsequent analysis tasks.

3.1.1 Data Loading and Initial Inspection

The original Road Accident Risk Dataset contains over 500,000 samples, which presents computational challenges for analysis and model training. To ensure efficient processing while maintaining statistical representativeness, we implemented a systematic sampling approach that selects the first 1,000 samples for our analysis.

Dataset Loading Strategy

Sampling Justification:

- *Computational Efficiency:* 1,000 samples provide sufficient data for meaningful analysis while ensuring reasonable processing times
- *Prototype Development:* Enables rapid iteration and model development before scaling to larger datasets

```
1 df_train = pd.read_csv('train.csv')
2 print(df_train.shape)
3 # Output: (517754, 14)
4 df_train.drop(['id', 'num_lanes', 'time_of_day'], axis=1, inplace=True)
5 df_train_sample = df_train.head(1000).copy()
6 print(df_train_sample.shape)
7 #output:df_train_sample shape: (1000, 11)
```

Listing 1: Dataset Loading and Initial Sampling

The sampling approach ensures we have a manageable yet representative subset of the original dataset. This 1,000-sample dataset maintains the essential characteristics of the full dataset while enabling efficient exploration, preprocessing, and model development.

3.2 Adding Missing value:

To simulate real-world data imperfections, we intentionally introduced missing values into specific attributes of the dataset. This process involved randomly removing a certain percentage of entries from selected features, thereby creating a more challenging and realistic dataset for analysis.

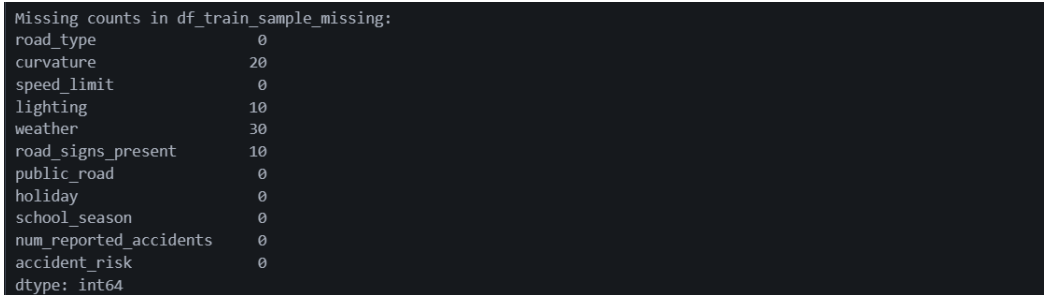
```
1 # create copies so original dfs remain available
2 df_train_missing = df_train.copy()
3 df_train_sample_missing = df_train_sample.copy()
4
5 # reproducible random selection
6 rng = np.random.RandomState(42)
7
8 missing_config = {
9     'curvature': 0.02,          # 2% missing (numeric)
10    'lighting': 0.05,           # 5% missing (categorical)
```

```

11     'weather': 0.03,          # 3% missing (categorical)
12     'road_signs_present': 0.01, # 1% missing (boolean -> will become
    object/float after NaN)
13     'lighting': 0.01,        # 1% missing (integer)
14 }
15
16 # helper to inject missing values into a dataframe
17 def inject_missing(df, config, rng):
18     n_rows = len(df)
19     for col, frac in config.items():
20         n_missing = int(np.floor(frac * n_rows))
21         if n_missing <= 0:
22             continue
23         idx = rng.choice(df.index, size=n_missing, replace=False)
24         df.loc[idx, col] = np.nan
25     return df
26
27 df_train_sample_missing = inject_missing(df_train_sample_missing,
    missing_config, np.random.RandomState(7))
28 print("\nMissing counts in df_train_sample_missing:")
29 print(df_train_sample_missing.isnull().sum())

```

Listing 2: Introducing Missing Values into the Dataset



```

Missing counts in df_train_sample_missing:
road_type          0
curvature          20
speed_limit        0
lighting           10
weather            30
road_signs_present 10
public_road         0
holiday            0
school_season      0
num_reported_accidents 0
accident_risk       0
dtype: int64

```

Figure 4: Missing value injection results

3.3 EDA :

Exploratory Data Analysis (EDA) was conducted on the dataset to uncover patterns, trends, and relationships within the data. This process involved visualizing the distribution of key features, examining correlations, and identifying potential anomalies or outliers.

- **Feature Distributions:** Histograms and bar plots were created to visualize the distribution of numerical and categorical features, respectively. This helped identify skewness, modality, and potential outliers.

```

1     # Plot numerical columns in a single row
2     numerical_cols = ['curvature', 'speed_limit', '
    num_reported_accidents']
3     n = len(numerical_cols)
4     fig, axes = plt.subplots(1, n, figsize=(5 * n, 5))
5     # Ensure axes is iterable

```

```

6     axes = np.atleast_1d(axes)
7
8     for i, col in enumerate(numerical_cols):
9         sns.histplot(df_train_sample_missing[col], bins=30, kde=
10             True, ax=axes[i])
11         axes[i].set_title(f'Distribution of {col}')
12         axes[i].set_xlabel(col)
13         axes[i].set_ylabel('Count')
14
15 plt.tight_layout()
16 plt.show()

```

Listing 3: Exploratory Data Analysis (EDA) Implementation

```

1     # Plot categorical columns in a single row
2     categorical_cols = ['road_type', 'lighting', 'weather', '
3         road_signs_present']
4     n = len(categorical_cols)
5     fig, axes = plt.subplots(1, n, figsize=(5 * n, 5))
6     # Ensure axes is iterable
7     axes = np.atleast_1d(axes)
8
9     for i, col in enumerate(categorical_cols):
10         sns.countplot(x=df_train_sample_missing[col], ax=axes[i])
11         axes[i].set_title(f'Distribution of {col}')
12         axes[i].set_xlabel(col)
13         axes[i].set_ylabel('Count')
14
15 plt.tight_layout()
16 plt.show()

```

Listing 4: Categorical Feature Distribution Visualization

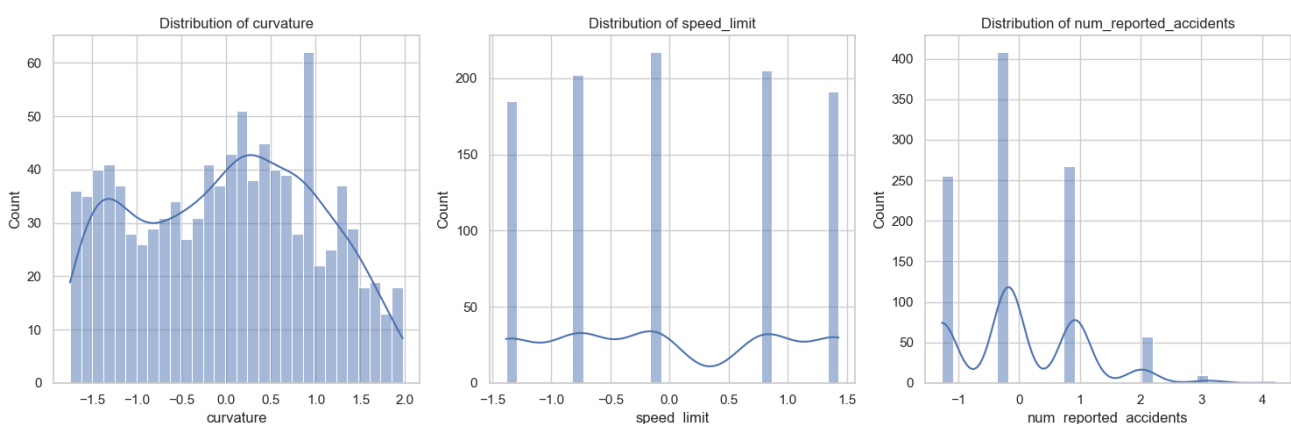


Figure 5: Feature distributions of key attributes(Numerical)

Findings from Numerical Distributions:

- *curvature*: The distribution is heavily right-skewed, with the vast majority of road segments having very low curvature (close to 0).

- *num_reported_accidents*: This is a discrete variable, with a decreasing frequency as the number of accidents increases.
- *speed_limit*: This plot clearly shows distinct, discrete peaks corresponding to standard speed limits (e.g., 35, 45, 60, 70 mph/kph).

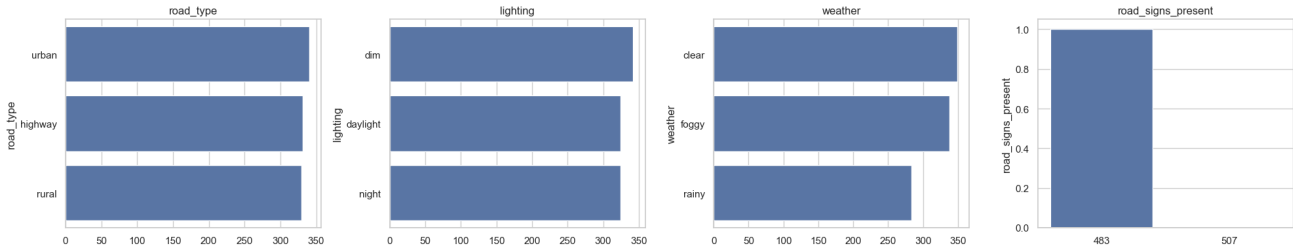


Figure 6: Feature distributions of key attributes(Categorical)

Findings from Categorical Distributions:

- *road_type*: The dataset contains three types, with 'urban' being the most frequent, followed by 'highway' and 'rural'.
- *lighting & weather*: For lighting, 'daylight' is the most common condition. For weather, 'clear' is the most common.
- *Boolean Features*: The boolean plots show the balance of *public_road* (mostly True), *holiday* (mostly False), *school_season* (roughly balanced), and *road_signs_present* (roughly balanced).